

VERIFYING SEMANTIC PREDICTIONS WITH HELD-OUT LiDAR OBSERVATIONS

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ABSTRACT

Semantic segmentation selects the most plausible known class for each LiDAR point. Recognizing an unfamiliar object requires an additional judgment: whether that class provides a convincing explanation of the observation. We introduce class-conditional observation verification, an end-to-end model that learns both judgments from normal semantic supervision. A full-scan encoder represents the observed point, while a separate context encoder predicts its return distribution under each known class. The predictor excludes all returns in the target angular cell, so its predictions are formed from surrounding measurements before the target range is evaluated. Each class combines a multimodal appearance representation with a multimodal distribution over the held-out log range. Their joint support determines both the semantic label and the unknown score. Normal labels train this joint class competition, and a normalized likelihood objective ties the predicted distributions to physical measurements. This formulation makes observation compatibility part of learning a semantic explanation. Evaluation on nuScenes and STU couples normal-class segmentation, real-anomaly ranking, high-recall false positives, and single-scan inference cost.

[Experimental findings.]

1 INTRODUCTION

A semantic label is a claim about the scene. When a LiDAR model labels a group of points as a car, it assigns those measurements a familiar interpretation. An unfamiliar road obstacle may nevertheless receive the same label because car is the closest available category. The distinction between the *best known explanation* and a *well-supported explanation* matters whenever a perception system encounters objects beyond its training vocabulary. STU makes this problem concrete through point-level annotations of unexpected objects in real road scenes (Nekrasov et al., 2025).

Learning this distinction from normal observations has practical appeal: familiar classes are extensively annotated, whereas future unknown objects cannot be enumerated. Normal feature distributions already provide useful rejection signals. LIDO organizes LiDAR features around semantic prototypes (Mosco et al., 2026); PixOOD models multiple normal feature configurations (Vojř et al., 2024); and RbA identifies image regions rejected by known classes (Nayal et al., 2023). These approaches establish a productive connection between semantic perception and unknown detection. The remaining question is what evidence can challenge a familiar-looking representation.

LiDAR provides a direct source of such evidence. A class hypothesis constrains how a surface can continue through a local scene. Nearby returns, their angular arrangement, and a candidate class jointly constrain the range of a return that has been withheld. For example, a road interpretation predicts a local continuation of the ground. A measured return that is incompatible with that continuation weakens the road interpretation. A vehicle interpretation can still support a normal vehicle in the same region. Unknown scoring should therefore depend on whether *any single known class* can account for both appearance and the measured return.

This reasoning motivates a particular use of prediction. We predict a physical measurement from its surroundings, evaluate the prediction at the actual measurement, and use that compatibility to learn the point’s semantic class. Class-conditioned reconstruction already supports open-set recognition and segmentation (Oza & Patel, 2019; Nunes et al., 2022), while semantic representations

guide multi-category anomaly reconstruction (Kim et al., 2026). Our formulation brings a held-out measurement likelihood directly into dense semantic class competition. Its prediction target is the recorded range, which remains fixed as the representation learns.

The model has two complementary information paths. A full-scan semantic encoder preserves the observed geometry and point details. A lightweight context encoder excludes the target angular cell and produces a return distribution for every known class. The same class vectors organize appearance modes and query the context. For each point and class, appearance support is multiplied by measurement support. Maximizing this joint support gives the normal label; the strength of that maximum gives the unknown score. Training on normal labels makes the measurement predictor responsible for choosing the correct semantic explanation, while likelihood training constrains the distributions it proposes.

The contribution is this connection between *what a class predicts* and *how that class is selected*. It has three concrete elements: a target-excluded, class-conditioned predictive distribution over raw LiDAR returns; a common class-support function for semantic labeling and unknown scoring; and normal-data learning that couples predictive likelihood with class competition. The empirical question is whether that connection preserves normal semantic quality while recovering unknown points that a conventional semantic model recognizes with high confidence. We evaluate these outcomes together, including false positives at high anomaly recall and the computation required for a complete scan.

2 RELATED WORK

Unknown recognition within semantic perception. Confidence and energy scores use information already available in a classifier (Hendrycks & Gimpel, 2017; Liu et al., 2020). Joint energy models connect classification and generative learning through shared class energies (Grathwohl et al., 2020). Dense open-set methods develop related ideas through class rejection, feature distributions, and joint discriminative and generative objectives (Nayal et al., 2023; Vojř et al., 2024; Grcić et al., 2022). In LiDAR, LIDO learns compact normal-class features (Mosco et al., 2026), and hierarchical Gaussian mixtures model class-conditional feature uncertainty (Shojaei Miandashti & Brenner, 2025). Other approaches train with generated or auxiliary unknown observations, including resized objects, diverse synthetic objects, diffusion-generated examples, and distribution-prior learning (Cen et al., 2022; Xu et al., 2025; Oh et al., 2026; Li et al., 2026a;b). These supervision choices define different experimental settings. Our training uses normal semantic labels, with the additional supervision supplied by the measured returns themselves.

Class-conditioned reconstruction and anomaly detection. Image resynthesis tests a semantic prediction through a generated image (Lis et al., 2019). C2AE learns class-conditioned reconstruction for open-set recognition (Oza & Patel, 2019); CoReSeg extends conditional reconstruction to dense labeling and uses the smallest class-conditioned reconstruction error to reject unknown pixels (Nunes et al., 2022). UniAD studies reconstruction of normal features and identifies the importance of controlling information transfer (You et al., 2022). In 3D, IMRNet detects object anomalies through iterative masked reconstruction (Li et al., 2024), while SeDiR uses object-category semantics and geometric attention to organize multi-category feature reconstruction (Kim et al., 2026). Our method predicts a held-out raw measurement for each class and places its likelihood inside the supervised pointwise class decision. The distinction concerns the prediction target, the information available to the predictor, and the role of verification in semantic learning.

Context prediction and probabilistic support. Blind-spot denoising learns from targets excluded from a predictor’s input (Batson & Royer, 2019; Laine et al., 2019; Krull et al., 2020). Conditional neural processes learn distributions at query locations from observed context (Garnelo et al., 2018; Kim et al., 2019). Point-cloud pretraining uses related prediction objectives to learn representations (Pang et al., 2022; Huang et al., 2023; Wu et al., 2025), and probabilistic LiDAR fields model range distributions for mapping (McDermott & Rife, 2024). We use local conditional prediction during both training and inference, with known classes as candidate explanations. Proper logarithmic scoring constrains the predictive distributions (Gneiting & Raftery, 2007). Studies of generative likelihood show that observation modeling and unknown recognition are distinct statistical questions (Nalisnick et al., 2019; Le Lan & Dinh, 2021). Our decision function combines class-specific ap-

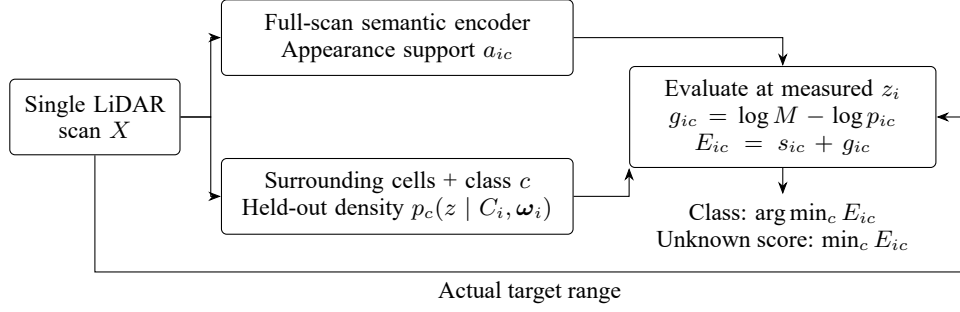


Figure 1: One class must support both the observed appearance and the held-out measurement. The semantic encoder reads the complete scan. The predictor reads independently encoded surrounding cells and excludes the target cell. Normal class supervision acts on the joint energy, connecting observation prediction to semantic learning.

pearance and conditional measurement support, and its evaluation includes both normal semantics and real unknowns.

3 CLASS-CONDITIONAL OBSERVATION VERIFICATION

3.1 SEMANTIC DECISIONS FROM JOINT CLASS SUPPORT

Let a scan be $X = \{(\mathbf{x}_i, I_i)\}_{i=1}^N$, where $\mathbf{x}_i \in \mathbb{R}^3$ is a return in the scan coordinate frame and I_i is its intensity. Define range $r_i = \|\mathbf{x}_i\|_2$, direction $\omega_i = \mathbf{x}_i/r_i$, and log range $z_i = \log r_i$. The known vocabulary is $\mathcal{C} = \{1, \dots, C\}$. Training provides a known class y_i , or an allowed set $Y_i \subseteq \mathcal{C}$ when a source label has several compatible interpretations. The output is a class $\hat{y}_i \in \mathcal{C}$ and a continuous unknown score u_i for every return.

For each candidate class, we compute appearance support $a_{ic} \in (0, 1]$ and a conditional density $p_{ic} = p_\theta(z_i | C_i, \omega_i, c)$, where C_i contains surrounding returns and excludes the target angular cell. A common upper bound M on the density defines measurement support $b_{ic} = p_{ic}/M$. The joint support and corresponding energy are

$$v_{ic} = a_{ic}b_{ic}, \quad E_{ic} = -\log v_{ic} = s_{ic} + g_{ic}, \quad s_{ic} = -\log a_{ic}, \quad g_{ic} = \log M - \log p_{ic}. \quad (1)$$

Both outputs follow from these class energies:

$$\hat{y}_i = \arg \min_{c \in \mathcal{C}} E_{ic}, \quad u_i = \min_{c \in \mathcal{C}} E_{ic}. \quad (2)$$

The label identifies the best-supported class. The unknown score retains its absolute support even when that class wins by a large margin over other classes.

A common class must explain both observations. The class index is shared before the final maximization:

$$\max_c (a_{ic}b_{ic}) \leq (\max_c a_{ic})(\max_c b_{ic}). \quad (3)$$

The right-hand side may combine the appearance of one class with the geometry of another. Equation (1) requires the two supports to agree on a class. For example, supports (0.9, 0.1) and (0.1, 0.9) give joint support 0.09, whereas independently choosing each maximum gives 0.81. This elementary property identifies the decision being learned: compatibility of a complete class explanation.

3.2 MULTIMODAL APPEARANCE SUPPORT

A LitePT-S backbone (Yue et al., 2026) encodes the scan after 5 cm voxel aggregation. Each original point also receives a detail embedding of its coordinates, intensity, and offset from its voxel center. Concatenating the decoded voxel feature and point detail produces $\mathbf{h}_i \in \mathbb{R}^D$, with $D = 48$. Thus points sharing a voxel retain distinct evidence.

Each class has a vector $\mathbf{q}_c$ and $A = 4$ learned appearance offsets δ_{cm} . The centers and support are

$$\mathbf{t}_{cm} = \sqrt{D} \frac{\mathbf{q}_c + \delta_{cm}}{\|\mathbf{q}_c + \delta_{cm}\|_2}, \quad a_{ic} = \frac{1}{A} \sum_{m=1}^A \exp\left(-\frac{\|\mathbf{h}_i - \mathbf{t}_{cm}\|_2^2}{D\tau}\right), \quad (4)$$

where $\tau = 0.2$. Equal center norms, temperature, and mode counts provide a common scale across classes. Multiple centers represent different normal configurations within a class. The same $\mathbf{q}_c$ conditions the measurement predictor, linking the class’s appearance representation to its predicted geometry.

3.3 PREDICTING RETURNS FROM TARGET-EXCLUDED CONTEXT

Information available to the predictor. We partition azimuth and elevation into nonoverlapping 0.5° cells. A pointwise encoder maps each return’s coordinates, intensity, and direction to a feature. Mean and maximum pooling within each cell produce an independent cell token. For a target cell, the context consists of eight neighboring offsets at each of two radii, one and three cells. Its own token is excluded. Class queries attend to these context tokens through two shared attention layers of width 48.

This construction fixes the information boundary before cross-cell interaction. Target range, target intensity, and target return count do not enter the predictive state. The query contains the target cell’s angular position and the candidate class. The full-scan semantic features follow their own path in Figure 1. At inference, all cells undergo the same local exclusion while sharing the point encoder and context projections.

A distribution of plausible normal returns. For each target cell j and class c , the predictor outputs $K = 3$ mixture weights w_{jck} , mean parameters, and scales σ_{jck} . Let (d_a, d_e) be a ray’s azimuth and elevation offsets from the cell center, normalized by the half-cell width, and define

$$\phi_i = (d_a, d_e, d_a^2, d_a d_e, d_e^2), \quad \mu_{ick} = \bar{z}_{C_j} + \alpha_{jck} + \beta_{jck}^T \phi_i. \quad (5)$$

Here $\bar{z}_{C_j}$ averages the mean log ranges of available neighboring cells. Learned quadratic coefficients express local range variation across rays. The scale is $\sigma_{jck} = \sigma_{\min} + \text{softplus}(\rho_{jck})$, with $\sigma_{\min} = 10^{-3}$, and mixture weights sum to one within each class.

The conditional density is a Student- t mixture with three degrees of freedom:

$$p_{ic} = \sum_{k=1}^K w_{jck} \frac{2}{\pi \sqrt{3} \sigma_{jck}} \left(1 + \frac{(z_i - \mu_{ick})^2}{3\sigma_{jck}^2}\right)^{-2}. \quad (6)$$

The mixture represents alternative normal surfaces, and its learned scale represents the spread of their possible observations. These parameters are predicted from context. The held-out measurement is used when evaluating Equation (6). Appearance modes and return components express different variations and have separate indices.

Measurement support includes predictive concentration. Each component is bounded above by $M = 2/(\pi \sqrt{3} \sigma_{\min})$, so the mixture yields $0 < b_{ic} \leq 1$. Equation (1) retains the scale normalization of the density. At a component mean, increasing its scale by a factor of ten raises its negative log density by $\log 10$. A broad prediction therefore carries a cost as well as accommodating a wider range of returns. For a completely isolated target cell, we set $b_{ic} = 1$ for all classes, giving the appearance-based decision. The predicted variable is the log range of an existing return, with density defined relative to $d(\log r)$.

3.4 LEARNING VERIFICATION THROUGH NORMAL CLASS COMPETITION

The class distribution used for semantic learning is

$$P_\theta(c \mid X, i) = \frac{\exp(-E_{ic})}{\sum_{d \in \mathcal{C}} \exp(-E_{id})}, \quad \ell_{\text{joint}, i} = -\log \sum_{c \in Y_i} P_\theta(c \mid X, i). \quad (7)$$

For a fine label $Y_i = \{y_i\}$ with available neighboring cells, normal semantic supervision trains the observed appearance and the measurement prediction to favor the same class. In particular,

$$\frac{\partial \ell_{\text{joint},i}}{\partial \log p_{ic}} = P_\theta(c \mid X, i) - \mathbf{1}[c = y_i]. \quad (8)$$

Thus the true class is encouraged to explain the measured return, and competing known classes are compared through the same evidence. This gradient is the direct link between observation verification and semantic learning.

The predictor also receives a normalized likelihood objective. Its context states produce class logits η_{ic} , which define a distribution restricted to the allowed label set,

$$\pi_{ic}^Y = \frac{\exp \eta_{ic}}{\sum_{d \in Y_i} \exp \eta_{id}} \quad (c \in Y_i), \quad \ell_{\text{pred},i} = -\log \sum_{c \in Y_i} \pi_{ic}^Y p_{ic}. \quad (9)$$

For a fine label, this is the negative log likelihood under the true class. The logarithmic score rewards predictive distributions that fit the normal observations (Gneiting & Raftery, 2007). A context-class loss supervises the total probability of Y_i . An appearance-only class loss, obtained by replacing E with s in Equation (7), supplies semantic supervision to every reliably labeled point. A compactness loss strengthens absolute appearance support,

$$\ell_{\text{compact},i} = -\log \left(\frac{1}{|Y_i|} \sum_{c \in Y_i} a_{ic} \right). \quad (10)$$

The training objective is

$$\mathcal{L} = \mathcal{L}_{\text{joint}} + 0.5\mathcal{L}_{\text{appearance}} + 0.05\mathcal{L}_{\text{compact}} + 0.2\mathcal{L}_{\text{pred}} + 0.2\mathcal{L}_{\text{context}}. \quad (11)$$

We balance classification losses over the distinct allowed label sets present in a frame. Predictive likelihood and compactness are point averages. Up to 4,096 normally labeled points per frame supervise joint verification, while the appearance loss covers all reliable normal labels. The complete scan supplies the input. Appendix B specifies these reductions and the source-to-target schedule.

Inference and a normal reference scale. A single scan supplies the semantic features and all context predictions. Class parameters are shared across angular cells, and neighboring tokens are encoded once. Each class density is evaluated at the original point’s range, yielding Equation (2). A common strictly increasing transform fitted to normal development scores expresses u_i relative to the upper tail of normal class energy. Its role is a common reference scale for the same underlying ranking.

4 EXPERIMENTS

The evaluation links three questions: whether observation verification maintains accurate known-class segmentation, whether it recovers unfamiliar points confidently assigned to known classes, and whether those gains remain useful at high anomaly recall.

4.1 DATA, TRAINING, AND EVALUATION

Normal supervision. nuScenes (Caesar et al., 2020) supplies 28,130 training scans from 700 scenes and a scene-disjoint development set of 6,019 scans from 150 scenes. STU sequence 206 supplies 449 target-domain normal training scans; sequence 201 supplies 682 normal development scans. Training uses trusted normal labels within 2.5–50 m, while all recorded returns remain available as inputs. Coarse nuScenes labels specify sets of compatible STU classes. Fine source annotations refine selected bicycle, parking, and building points. Ambiguous points carry no semantic loss. The vocabulary contains the 19 STU normal classes.

Optimization. The backbone starts from official nuScenes-pretrained LitePT-S weights. All other parameters are initialized randomly. Source training comprises two passes over nuScenes, followed by eight passes over sequence 206 with one source replay scan per four target scans. AdamW uses

Table 1: Normal perception and real-anomaly segmentation. All metrics are percentages. The first group uses matched normal data and training budgets; dashes reserve pending results. Published results in the second group retain their original training resources; n/r denotes an unreported normal-semantic result.

Model	201 mIoU↑	STU validation			STU test		
		AP↑	AUROC↑	FPR95↓	AP↑	AUROC↑	FPR95↓
Appearance only	—	—	—	—	—	—	—
Separate objectives	—	—	—	—	—	—	—
Joint verification	—	—	—	—	—	—	—
MC dropout (Nekrasov et al., 2025)	n/r	0.17	65.76	79.82	0.11	61.51	82.37
RbA (Nekrasov et al., 2025)	n/r	1.64	73.00	100.00	0.81	66.38	100.00
MaxLogit (Nekrasov et al., 2025)	n/r	2.02	87.27	68.76	0.95	84.53	81.49
Deep ensemble (Nekrasov et al., 2025)	n/r	6.94	90.93	37.34	5.17	86.74	58.05
LIDO (Mosco et al., 2026)	n/r	27.53	95.05	34.86	14.99	93.67	34.29

an effective batch size of two, weight decay 0.005, 3% linear warmup, and cosine decay to 5% of the peak rate. The source peak rates are 10^{-4} for the backbone and 8×10^{-4} for new parameters; target rates are 3×10^{-5} and 2×10^{-4} . Selection uses normal-development mean intersection-over-union (mIoU), with the normal objective breaking ties. The independent controls use the same initial weights, data order, augmentations, and update budget.

Evaluation. Normal perception is evaluated by per-class IoU and mIoU on sequence 201, retaining the set of classes represented in its ground truth. STU’s real-anomaly validation and hidden test sets supply average precision (AP), area under the receiver operating characteristic curve (AUROC), and the normal false-positive rate at 95% anomaly recall (FPR95). Evaluation follows the official distance, label, and anomaly-instance rules (Nekrasov et al., 2025). The official anomaly-evaluation mask determines the normal denominator. Normal training labels and anomaly-evaluation labels serve their respective tasks. We also examine recall at matched normal false-positive rates of 1% and 5%, and report full-scan latency and peak memory on the same device.

4.2 NORMAL PERCEPTION AND REAL-ANOMALY SEGMENTATION

Table 1 places normal semantic accuracy beside the three anomaly metrics. The controlled comparison uses an independently trained appearance model and an equally trained joint model with the same backbone and normal data. Published STU methods provide an external comparison, with training resources and auxiliary supervision recorded explicitly. This arrangement connects unknown recognition to the quality of the underlying normal perception model.

The published LIDO model uses MinkowskiNet trained on SemanticKITTI (Mosco et al., 2026). The STU implementations of MC dropout, RbA, MaxLogit, and deep ensembles use Mask4Former-3D trained on SemanticKITTI, Panoptic-CUDAL, and the STU normal training split (Nekrasov et al., 2025). These rows report the papers’ original results. The matched-data comparison uses the nuScenes and STU normal training described above.

4.3 EFFECT OF OBSERVATION EVIDENCE ON SEMANTIC DECISIONS

The central comparison removes observation compatibility from the supervised class competition while retaining the predictive objective, shared class parameters, and joint inference score. The difference between this control and the complete model measures the effect of learning to classify with observation evidence. An independently trained appearance-only model establishes the normal semantic reference.

For unknown points, the reference classifier defines a fixed subset whose maximum class probability is at least 0.9. Recall on this subset measures recovery of confident semantic misrecognitions. At each matched false-positive rate, paired point identities determine how many unknowns are recovered or lost. On normal points, the same analysis counts corrected and newly introduced semantic

Normal-class IoU changes and unknown recall at matched false-positive rates

Figure 2: Effect of verification on semantic decisions and unknown recognition. Left: per-class IoU relative to the independently trained appearance model. Right: recall of confidently misrecognized unknown points at matched normal false-positive rates. The same reference model defines the subset for every method.

Table 2: Changes in pointwise decisions relative to the appearance model. Corrected and introduced normal errors are percentages of all evaluated normal semantic points. Unknown recall is measured on the fixed subset confidently assigned a normal class by the reference model, at matched normal false-positive rates.

Model	Normal semantic errors		Unknown recall↑	
	Corrected↑	Introduced↓	at 1% FPR	at 5% FPR
Appearance only	—	—	—	—
Separate objectives	—	—	—	—
Joint verification	—	—	—	—

errors, with per-class IoU exposing the distribution of these changes. Table 2 and Figure 2 join these two views of observation verification.

4.4 PREDICTIVE QUALITY, SPARSE RETURNS, AND COMPUTATION

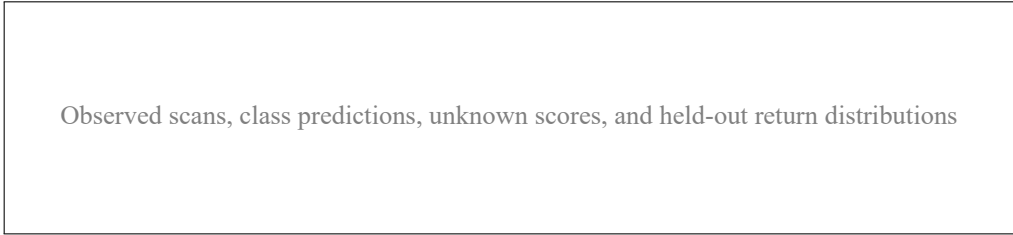
Normal-data diagnostics examine held-out log likelihood, the coverage and width of 90% predictive intervals, and the use of mixture components. Jointly examining coverage and width tests whether the model accommodates normal variation with informative predictions (Gneiting & Raftery, 2007). Stratification by range and angular-cell return count connects these quantities to sparse observations. The anomaly analysis uses the same strata to locate changes in missed unknown points and normal false positives.

Qualitative examples place the scan, normal labels, unknown scores, and class-conditioned return distributions side by side. The most informative cases compare a normal and an unfamiliar object with similar appearance-based class confidence. The plots show which candidate class remains supported by the actual measurement. Timing includes scan preparation, the semantic encoder, contextual prediction, and pointwise scoring, with peak memory measured under the same execution conditions for each model.

5 CONCLUSION

We formulate normal semantic perception as a class decision supported by both appearance and a testable prediction of the observed return. A context branch predicts the held-out log range for every known class, and its likelihood enters the same class competition that learns semantic segmentation. The resulting class energy provides both a normal label and an unknown score. This formulation gives normal labels and physical measurements complementary roles in learning when a familiar explanation is supported.

[Experimental findings.]



Observed scans, class predictions, unknown scores, and held-out return distributions

Figure 3: Class explanations at observed returns. Each selected point is accompanied by the predicted return distributions for the most relevant known classes and its measured range. Normal sparse objects and confidently misrecognized unknown objects are presented with the same visualization scale.

AI USE STATEMENT

We used OpenAI Codex for literature retrieval and synthesis, research-question refinement, model formulation, mathematical derivations, experiment design, data processing and annotation review, implementation, debugging, analysis of development results, and manuscript preparation. Assistance with the manuscript included drafting, editing, reference verification, diagrams, and typesetting. Source texts were consulted for method comparisons; the formulation was checked against the implementation; numerical and real-scan checks examined the implemented computations. The authors are responsible for the submitted text, claims, code, and data artifacts.

REPRODUCIBILITY STATEMENT

Section 3 defines the class energies and training objectives. Appendix A specifies the context construction and predictive distribution, Appendix B gives supervision and optimization details, and Appendix C defines the evaluation reductions. The experiments use nuScenes and STU with the stated scene and sequence assignments.

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A PREDICTIVE ARCHITECTURE AND DECISION FUNCTION

A.1 ANGULAR CONTEXT AND POINT IDENTITY

The scan origin defines an effective direction for every recorded return. Azimuth is periodic, and elevation is bounded. Cell assignment depends only on direction. Each cell token concatenates the mean and maximum of independently encoded point features and projects the result to 48 dimensions. A positional embedding uses the cell-center azimuth and elevation. Neighbor offsets are $(d_a, d_e) \in \{-1, 0, 1\}^2 \setminus \{(0, 0)\}$ at radii one and three cells. Missing cells are masked in attention. A learned empty token supplies a finite context state for an isolated cell; its measurement contribution in Equation (1) is then zero.

The class queries are normalized shared vectors, $\sqrt{48} \mathbf{q}_c / \|\mathbf{q}_c\|_2$, plus the target-cell positional embedding. Two cross-attention layers use three heads and a feed-forward hidden width of 96. Neighboring key and value projections are shared across queries. A linear output layer predicts three sets of eight parameters per class: one mean offset, five quadratic coefficients, one scale parameter, and one mixture-weight logit. A separate linear output produces the context class logits η_{ic} .

Each target point reads the prediction for its own cell and evaluates its own angular offset and range. Multiple returns in a cell share prediction parameters, with direction-dependent means. Output indices retain the original return identities throughout voxel aggregation, point decoding, and evaluation. Empty acquisition slots are restored only when writing the final arrays.

Table 3: Architecture constants.

Component	Setting
Semantic backbone	LitePT-S, official nuScenes pretraining
Voxel width	0.05 m
Point detail input	Position, intensity, within-voxel offset
Point embedding dimension	48
Known classes / appearance modes per class	19 / 4
Appearance temperature	0.2
Angular-cell width	0.5°
Context cells / attention layers / heads	16 / 2 / 3
Return components / degrees of freedom	3 / 3
Minimum log-range scale	0.001
Mean surface	Quadratic in normalized angular offsets

A.2 NORMALIZATION AND CLASS COUPLING

The unit-scale Student- t density with three degrees of freedom is

$$t_3(v) = \frac{2}{\pi\sqrt{3}}(1 + v^2/3)^{-2}. \quad (12)$$

For a location μ and scale σ , $\sigma^{-1}t_3((z - \mu)/\sigma)$ integrates to one over $z \in \mathbb{R}$. Nonnegative mixture weights summing to one preserve this property. Since $\sigma \geq \sigma_{\min}$, every component and its mixture are bounded by $M = 2/(\pi\sqrt{3}\sigma_{\min})$. Hence $g_{ic} \geq 0$ and $E_{ic} \geq 0$.

For supported cells, the class logits can be written $-s_{ic} + \log p_{ic} - \log M$. The common constant cancels from the normalized class probabilities. Differentiating their cross-entropy gives Equation (8). For an allowed set, let

$$P_{ic}^Y = \frac{\mathbf{1}[c \in Y_i] P_\theta(c | X, i)}{\sum_{d \in Y_i} P_\theta(d | X, i)}. \quad (13)$$

Then $\partial \ell_{\text{joint}, i} / \partial \log p_{ic} = P_\theta(c | X, i) - P_{ic}^Y$. The allowed set therefore receives joint support according to its currently compatible members. The predictive loss in Equation (9) uses a context-derived class mixture within that set.

The joint function v_{ic} is a discriminative class support. Its appearance encoder and context encoder both process the scan, with distinct information restrictions. The density p_{ic} describes the conditional log-range prediction. The class competition uses their combined support as a discriminative energy.

B NORMAL SUPERVISION AND OPTIMIZATION

B.1 FINE AND COARSE NORMAL LABELS

The target vocabulary is car, bicycle, motorcycle, truck, other vehicle, person, bicyclist, motorcyclist, road, parking, sidewalk, other ground, building, fence, vegetation, trunk, terrain, pole, and traffic sign. Each reliable target label supplies a singleton Y_i . Compatible source categories provide either a singleton or a set. For example, source vegetation may correspond to vegetation or trunk, and broad source ground labels may admit several target ground classes. This set supervision preserves the source annotation’s resolution.

Training source annotations additionally identify rider-associated bicycle points in 1,586 frames from 74 scenes, 328 verified parking points from three training scenes, and 1,328 building points from four training frames. These annotations refine the same source scans. Target sequence 206 has no fine parking or bicyclist supervision. Half of the source replay positions prioritize source frames with reliable labels for these classes, while the other half follow the ordinary source sequence. Source replay remains one scan per four target scans.

Source training and development scenes, logs, and scan identities are disjoint. Training and normal-development supervision uses the 2.5–50 m range. STU labels 0, 1, 2, 52, and 99 supply no fine normal-class supervision. The normal training and development sequences exclude scans carrying the target anomaly label. Intensity is divided by 255 for nuScenes and retains the recorded scale for STU.

B.2 LOSS REDUCTIONS AND SAMPLING

Let V be the reliably labeled points in a frame and $Q \subseteq V$ the selected joint queries. Queries first cover the represented normal labels and then fill the remaining quota, up to 4,096 points. Let $\mathcal{G}(S)$ be the distinct allowed label sets among points in S , and $S_G = \{i \in S : Y_i = G\}$. A class-set-balanced loss is

$$\mathcal{B}_S(\ell) = \frac{1}{|\mathcal{G}(S)|} \sum_{G \in \mathcal{G}(S)} \frac{1}{|S_G|} \sum_{i \in S_G} \ell_i. \quad (14)$$

Joint classification uses $\mathcal{B}_Q$, and appearance classification uses $\mathcal{B}_V$. The context-class loss applies the same reduction to context-supported queries. Predictive likelihood averages over supported queries; compactness averages over all queries. Input returns are retained independently of these supervision subsets.

B.3 TRAINING SCHEDULE AND MODEL SELECTION

Table 4: Training schedule and common optimization settings.

Setting	Source stage	Target stage
Data	nuScenes	STU 206 + source replay
Passes over primary training set	2	8
Training scan visits	56,260	4,488
Optimizer updates	28,130	2,244
Backbone peak learning rate	10^{-4}	3×10^{-5}
Other-parameter peak learning rate	8×10^{-4}	2×10^{-4}
Effective batch size	2	2
Development scans for selection	50	68
Selection interval (updates)	2,000	225

The source development subset samples one central frame from each of 50 independent scenes. The target subset samples 68 frames uniformly along sequence 201. Each selection step evaluates the normal semantic decision defined by the relevant model. The final evaluation covers all 682 normal development scans.

Optimization uses FP32, AdamW with $\epsilon = 10^{-6}$, gradient clipping at norm one, and rotation about the gravity axis. Gradient accumulation implements the effective two-scan batch. The training bud-

get is eight hours on an RTX 5080 Laptop GPU with 16 GB memory. The table specifies the planned complete schedule; actual completed updates and the selected update are reported with the experimental results.

C EVALUATION DETAILS

C.1 NORMAL SEMANTIC QUALITY

For the aggregate pointwise confusion matrix H , class c has

$$\text{IoU}_c = \frac{H_{cc}}{\sum_d H_{cd} + \sum_d H_{dc} - H_{cc}}. \quad (15)$$

The primary normal mIoU averages the classes with nonzero ground-truth count, fixed across models. A second statistic averages classes with nonzero union. Sequence 201 has no ground-truth points for motorcycle, other vehicle, bicyclist, motorcyclist, and other ground. Per-class tables retain these categories and their ground-truth counts.

For held-out prediction, normal diagnostics include the average negative log likelihood, median range error, and coverage and metric width of the 90% predictive interval. Fine-label points use their true class. Coarse labels use the allowed-class mixture. Full-point semantic statistics and sampled predictive diagnostics retain separate denominators.

C.2 UNKNOWN RANKING AND HIGH-RECALL FALSE POSITIVES

STU anomaly evaluation follows the official point mask, range limit, and minimum anomaly-instance visibility of five points. Normal points are those designated normal by that evaluator, including categories not used for fine semantic training. Scores are pooled over the full evaluated point set. AP integrates precision against recall increments; AUROC integrates the false-positive and true-positive curve. FPR95 follows the official first operating point whose true-positive rate exceeds 0.95. Equal scores form a single threshold group.

Let the independently trained appearance classifier have probabilities P^{app} . Confidently misrecognized unknown points form the common subset

$$U_{\text{conf}} = \{i : y_i = \text{unknown}, \max_c P^{\text{app}}(c | X, i) \geq 0.9\}. \quad (16)$$

At normal false-positive budgets of 1% and 5%, comparison uses common attainable false-positive counts, retaining tied-score groups. Recall is then evaluated on all unknown points and on U_{conf} . Paired point identities allow recovered and lost detections to be counted directly. These operating points describe the evaluation curves.

C.3 NORMAL REFERENCE SCORES AND OBSERVATION STRATA

A single normal reference distribution is fitted to $\min_c E_{ic}$ on sequence 201. Each scan contributes up to 2,048 uniformly sampled trusted normal points. The representation uses 129 quantile knots, with denser sampling in the upper tail. Tied knots are merged, and a common piecewise-linear mapping expresses scores on a negative log-tail scale. Positive-slope linear extrapolation extends both tails beyond the observed knots. If all knots coincide, an affine transform with positive slope is used. The same strictly increasing mapping applies to all classes and preserves the ordering of raw scores.

Distance strata are bounded by 10, 20, 30, and 40 m within the evaluation range. Angular-cell return-count strata are bounded by 1, 3, 7, and 15. Predictive statistics, semantic errors, and unknown detection outcomes use these common strata. Latency measurements cover scan preparation and the full model under a shared device configuration.

D ADDITIONAL RESULTS

Table 5: Full-scan computational cost and normal predictive quality. Entries reserve measurements under a common device and execution configuration.

Model	Latency (ms)↓	Peak memory (GB)↓	NLL↓	90% coverage
Appearance only	—	—		
Separate objectives	—	—	—	—
Joint verification	—	—	—	—

Range and return-count strata: normal false positives, anomaly recall, and predictive intervals

Figure 4: Observation-dependent behavior. Results are stratified by measured range and angular-cell return count, with point counts reported for each stratum.